BismillahirRahmanir Rahim

International Islamic University Chittagong

Department of Computer Science & Engineering

B. Sc. in CSE Final-term Examination, Autumn-2018

Course Code: EEE 1221 Course Title: Electronics

Total marks: ***50*** Time: **2 hour 30 minutes**

[Answer any *two* questions from **Group-A** and any *three* questions from **Group-B**; Separate answer script must be used for Group-A and Group-B.]

Group-A

[Answer any *two* questions]

a) What is FET? Write down the classification of FET. 2

b) What is JFET? Describe its construction and working principle (N-Channel JFET). 5

c) A JFET has the following parameters: I_{DSS}= 32 mA; $V_{GS(off)}$= -8V; V_{GS}= -4.5 V. Find out the value of drain current. 3

a) What is switching circuit? Write down the classification of it. 2

b) Explain the switching action of transistor with proper diagram in 3

i) OFF region

ii) ON region

c) Draw the drain or output Characteristics curves of Enhancement type of MOSFET. 2

d) A transistor is used as switch. If V_{CC}=10V; R_C= 1KΩ and I_{CBO}= 10 μA, determine the value of V_{CE} when the transistor is 3

(i) Cut off

(ii) Saturation

a) Draw the symbol of n- channel and p-channel enhancement type of MOSFET. 2

b) Explain the operation principle of Monostable Multivibrator with proper diagram. 5

c) An E-MOSFET gives $I_{D\,(on)}$ =500mA at V_{GS}=10V and $V_{GS\,(th)}$=1V. Determine the drain current for V_{GS}=5V. 3

Group-B

[Answer any *three* questions]

4.a) Define Operational Amplifier. Write down the Ideal Characteristics of an OP AMP.

b) Draw the circuit diagram of inverting and non-inverting OP-AMP and find the gain for both.

c) From the Op-amp circuit, if v_i =0.5 V, calculate (i) the output voltage v_o, and (ii) the current in the 10 KΩ resistor.

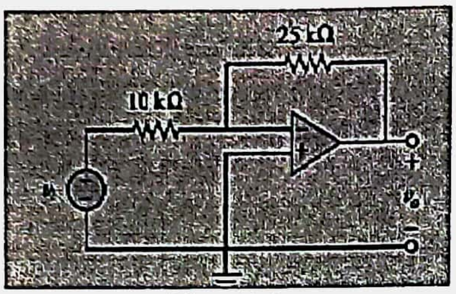


Fig. 4(c)

5.a) What is negative feedback?

b) Explain the principle of negative feedback in amplifier.

c) Derive the gain of negative feedback amplifier.

d) When negative voltage feedback is applied to an amplifier of gain 100, the overall gain falls to 50.

(i) Calculate the fraction of the output voltage feedback.

(ii) If this fraction is maintained, calculate the value of the amplifier gain required if the overall stage gain to be 75.

6.a) What is an electronic oscillator? Write down the name of different types of transistor oscillators.

b) With proper circuit diagram explain the operation of Hartley Oscillator.

c) In a Hartley oscillator the tank coil has two sections of inductance 80 mH and 20 mH. The capacitor has a capacitance of 500 pF. Neglecting the mutual inductance of the coil, find its frequency of oscillation.

7.a) Write short note on comparator circuit and characteristics of it.

b) Draw the circuit diagram of Peak Detector circuit.

c) With proper circuit diagram explain the operation of Schmit Tigger.

Bismillahir Rahmanir Rahim

International Islamic University Chittagong

Department of Computer Science & Engineering

B. Sc. in CSE Semester Final Examination, Autumn- 2018(Special)

Course Code: EEE- 1221 **Course Title: Electronics**

Total marks: *50* Time: 2 and half hours

[Answer any ***two*** questions from **Group-A** and any ***three*** questions from **Group-B**; Separate answer script must be used for Group-A and Group-B.]

Group-A

What is FET? How many types of FET are available? 2

What is JFET? Describe its construction and working principle (N-Channel JFET). 5

A JFET has the following parameters: I_{DSS}= 22 mA; $V_{GS(off)}$= -6V; V_{GS}= -3.5 V. Find out the value of drain current. 3

What is switching circuit? Write down the classification of it. 2

Explain the switching action of transistor with proper diagram in 4

i) OFF region

ii) ON region

A transistor is used as switch. If V_{CC}=10V; R_C= 1KΩ and I_{CBO}= 10 μA, determine the value of V_{CE} when the transistor is (i) cut off (ii) saturated. 4

Why FET is called Voltage-controlled device? Explain your answer with proper diagram. 2

What are the applications of monostable multivibrator? Explain its working principle. 4

What is electro-mechanical switch or relay. Explain the working principle of relay with proper diagram. 4

Group-B

What is operational amplifier? Write down the applications of operational amplifier. 2

Explain how an Op-Amp can be used as an integrator amplifier. 4

For Fig. 1, let R_f= 250 kΩ , R_i= 10kΩ, and E_i= - 0.5 V. Calculate : 4

i. I_o;

ii. The voltage across R_f;

iii. V_o

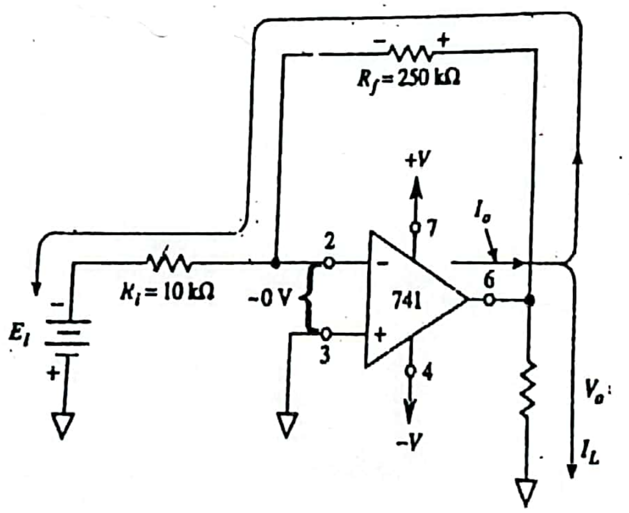


Fig-1

5.a) What is negative feedback?

b) Explain the principle of negative feedback in amplifier.

c) Show that the negative current feedback increases the output impedance of an amplifier.

d) When negative voltage feedback is applied to an amplifier of gain 100, the overall gain falls to 50.

   (i) Calculate the fraction of the output voltage feedback.

   (ii) If this fraction is maintained, calculate the value of the amplifier gain required if the overall stage gain to be 75.

6.a) What is an electronic oscillator? Write the different types of electronic oscillators.

b) Draw the circuit diagram of a Hartley Oscillator and derive an expression for the frequency of this circuit.

c) In a Hartley oscillator the tank coil has two sections of inductance 20 mH and 10 mH. The capacitor has a capacitance of 300 pF. Neglecting the mutual inductance of the coil, find its frequency of oscillation.

7.a) Draw the circuit diagram of inverting amplifier and derive an expression for its voltage gain.

b) Draw a circuit diagram of a Schmitt trigger and describe its principle of operation.

International Islamic University Chittagong

Department of Computer Science and Engineering

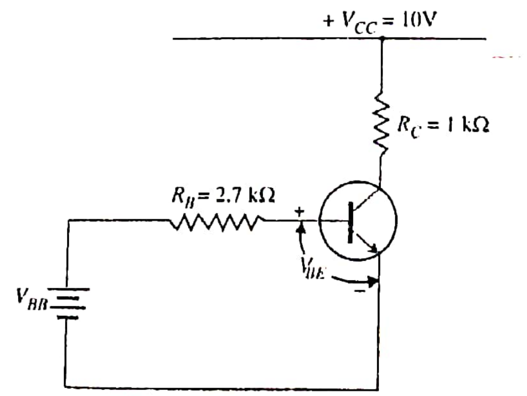
B.Sc. in CSE, Final Examination, Spring 2022

Course Code: **EEE-1221** Course Title: **Electronics**

Time: **2** hours **30** minutes Full Marks: **50**

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

		Part A [Answer all the questions from the followings]	
1.	a)	Write down the difference between JFET and BJT. Describe its construction and working principle of (N-Channel JFET). i) When gate-source voltage (V_{GS}) is applied and drain-source voltage is zero i.e. V_{DS}= 0V. ii) When drain-source voltage (V_{DS}) is applied at constant gate-source voltage (V_{GS})	CO4
		Or,	
		What is MOSFET? What are the different types of MOSFET? With a neat diagram, explain the working principle of an n-channel enhancement type MOSFET.	
1.	b)	A JFET has the following parameters: I_{DSS}= 32 mA; $V_{GS(off)}$= -8V; V_{GS}= -4.5 V. Find out the value of drain current.	CO4
2.	a)	What is a multivibrator? Mention different types of multivibrators with proper waveshapes. With neat diagrams, explain the working of an astable multivibrator. Or, What is an oscillator? What are the essentials of an oscillator? With the help of a neat diagram, describe the circuit operation of a Hartley oscillator.	CO4
2.	b)	Fig. 2(b) shows the transistor switching circuit. Given that R_B = 2.7 kΩ, V_{BB} = 2V, V_{BE} = 0.7V and V_{knee} = 0.7V. i) Calculate the minimum value of β for saturation. ii) If V_{BB} is changed to 1V and transistor has minimum β = 50, will the transistor be saturated.  Fig. 2(b)	CO4

Part B

[Answer the questions from the followings]

Q.		Question	CO	Level	Marks
3.	a)	Show that when the gain of summing amplifier is unity, the output voltage is the algebraic sum of the input voltages with proper circuit diagram. Or, Show that the output is the integral of the input with an inversion and scale multiplier of 1/RC.	CO5	A	6
3.	b)	Determine the output voltage from the circuit shown in Fig. 3(b) for each of the following input combinations:	CO5	A	4

V_1(V)	V_2(V)	V_3(V)
+ 10	0	+10
0	+10	+10
+10	+10	+10

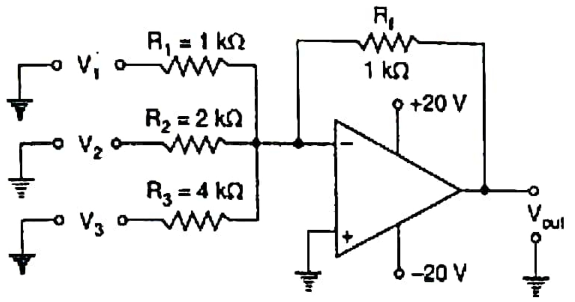
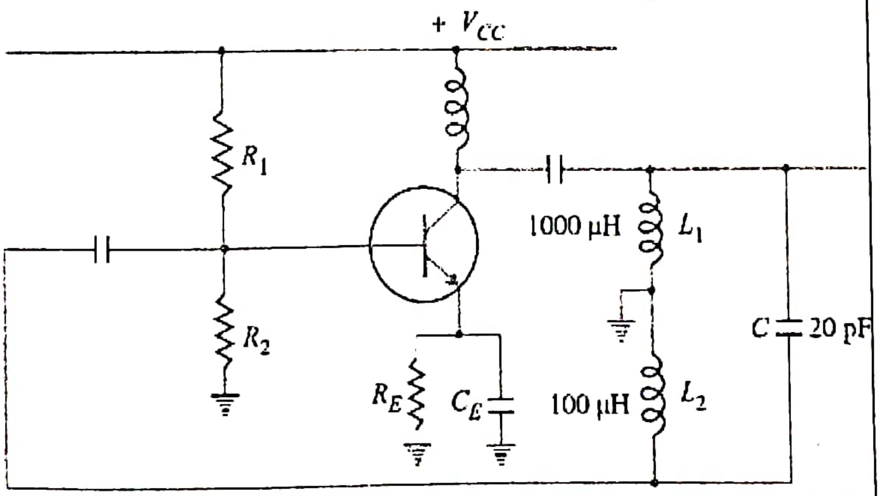


Fig. 3(b)

Q.		Question	CO	Level	Marks
4.	a)	What is negative feedback? Show that the input impedance of an amplifier increases due to negative feed.	CO5	U	6
4.	b)	Calculate the operating frequency and feedback fraction of the following oscillator where the mutual inductance between two coils is 20μH.	CO5	A	4



Q.		Question	CO	Level	Marks
5.	a)	What is negative feedback?. Show that the input impedance of an amplifier increases due to negative feed.	CO5	U	5
5.	b)	Write short on : i) Precision Rectifiers ii) Comparators Or, What is an operational amplifier? Draw the circuit diagram of non-inverting OP-AMP with indicating different terminals. Also show the voltage gain of a non-inverting amplifier is $1+\frac{R_f}{R_i}$.	CO5	U	5

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE

Final Exam, Autumn-2022

Course Code: **EEE-1221** Course Title: **Electronics**

Time: 2 hours 30 minutes Full Marks: 50

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

Course Outcomes (COs) of the Questions	
CO1	Describe the fundamentals of solid state electronics
CO2	Sketch the output wave-shape of different diode circuits
CO3	Differentiate the types of generated and filtered wave-shapes
CO4	Understand the basics of transistor and switching circuits
CO5	Analyze different operational amplifier circuits and their applications

Bloom's Levels of the Questions						
Letter Symbols	R	U	App	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Part A

[Answer the questions from the followings]

a) What is JFET? Draw the symbol of N-Channel JFET and P-Channel JFET. Describe the working principle of (N-Channel JFET), When gate-source voltage (V_{GS}) is applied and drain-source voltage is zero i.e. V_{DS}= 0V. CO4 U 5

OR,

With proper diagrams, describe the construction and working principle of an *n*-channel enhancement-type MOSFET.

b) Sketch the transfer and drain characteristics of n-channel enhancement type of MOSFET if V_T=4V and k=0.5 X 10^3 A/V^2. CO4 A 5

a) Describe the switching action of the transistor by showing the 'OFF' region, 'ON' region, and 'Active' regions on its output characteristics. CO4 U 5

OR,

Suppose you have given two transistors with few other passive elements, design a Multivibrator having one stable state. Explain its operation when a square wave will generate as Output.

) Determine the minimum high input voltage (+V) required to saturate the transistor switch shown in Fig.2 (b). CO4 A 5

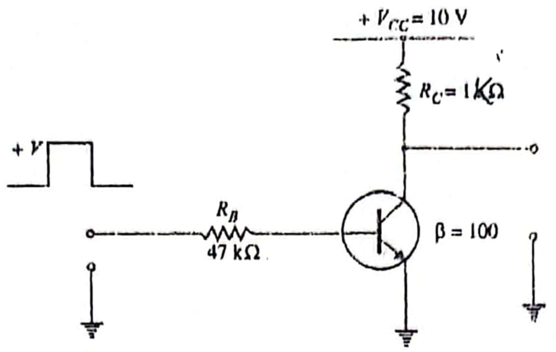


Fig. 2(b)

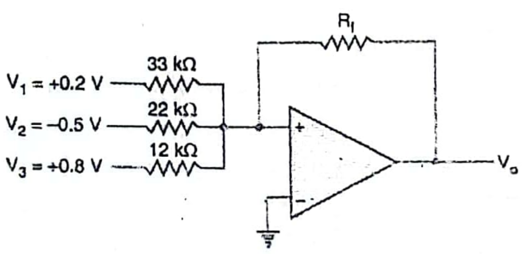
Part B

[Answer the questions from the followings]

3. a) What is an operational amplifier (OP-amp)? Draw the schematic symbol of an operational amplifier indicating the various terminals. CO5

3. b) Show the voltage gain of an inverting op-amp is equal to $-R_f/R_{in}$. CO5

3. c) If $R_f = 68\times10^3\ \Omega$, calculate the value of the output voltage of the following circuit: CO5



4. a) Describe the principle of a negative feedback amplifier and hence derive an expression for its gain. CO5

OR

Show that output is the differentiation of the input with an inversion and scale multiplier of RC.

4. b) Fig. 4(b) shows the square wave input to a differentiator circuit. Find the output voltage if input goes from 0V to 5V in 0.1 ms. CO5

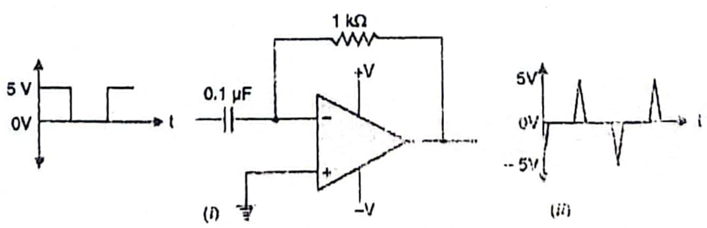


Fig.4(b)

5. a) What is an oscillator? With the help of a neat diagram, describe the circuit operation of a Hartley oscillator. CO5

OR

Show that for an op-amp with unity gain, the output voltage is the algebraic sum of the input voltages.

5. b) Write short on (i) Comparators and (ii) Precision rectifiers. CO5

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International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE

Final Exam, Spring-2023

Course Code: **EEE-1221** Course Title: **Electronics**

Time: 2 hours 30 minutes Full Marks: 50

The figures in the right-hand margin indicate full marks

Part A

[Answer the questions from the followings]

1. a) What is JFET? Draw the symbol of N-Channel JFET and P-Channel JFET. Describe the working principle of (N-Channel JFET). CO4 U 5
   i) When drain-source voltage (V_{DS}) is applied at constant gate-source voltage (V_{GS})

OR,

What are the main differences between enhancement-mode and depletion-mode MOSFETs? Explain the operation of n-channel enhancement mode MOSFET.

1. b) A JFET has a drain current of 5mA. If I_{DSS} =10 mA and $V_{GS(off)}$ = -6V, Find the value of i) V_{GS} and ii) V_P CO4 An 5

2. a) Describe the switching action of the transistor by showing the 'OFF' region, 'ON' region, and 'Active' regions on its output characteristics. CO4 U 5

OR,

Suppose you have given two transistors with few other passive elements, design a Multivibrator having no stable state. Explain its operation when a square wave will generate as Output.

2. b) Fig. 2(b) shows the transistor switching circuit. Given that R_B = 2.7 kΩ, V_{BB} = 2V, V_{BE} = 0.7V and V_{knee} = 0.7V. CO4 An 5
   i) Calculate the minimum value of β for saturation.
   ii) If V_{BB} is changed to 1V and transistor has minimum β = 50, will the transistor be saturated.

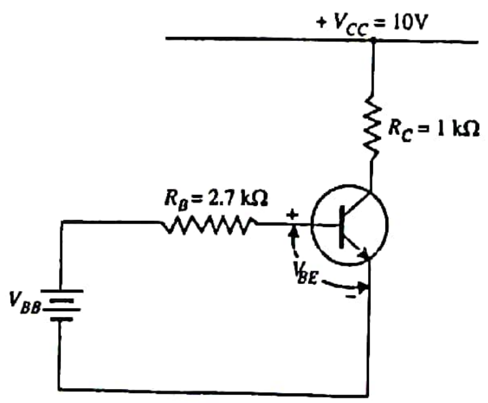

Fig. 2(b)

Part B

[Answer the questions from the followings]

3\. a) What is an operational amplifier (OP-amp)? Draw the schematic symbol of an operational amplifier indicating the various terminals. CO5

3\. b) Sketch a neat diagram and derive an expression for the voltage gain of a non-inverting amplifier. CO5

3\. c) Illustrate the output voltage waveform with proper mathematical expression for the circuit given below CO5

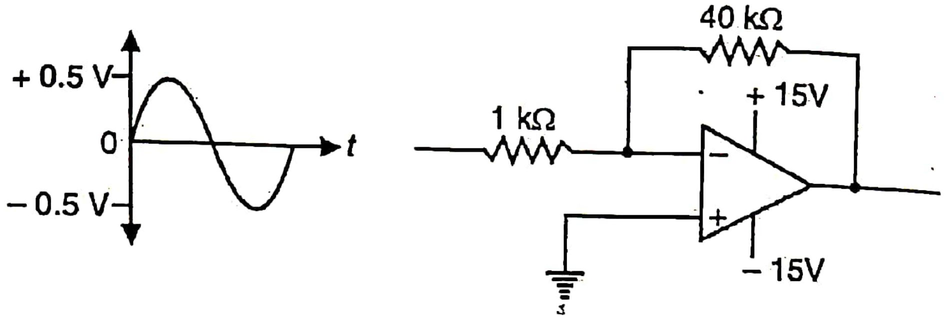

Fig. 3(c)

4\. a) What is negative feedback? Explain the principle of negative feedback in amplifier. CO5

OR

What is feedback? Explain the principle of negative feedback in amplifier.

4\. b) When negative voltage feedback is applied to an amplifier of gain 100, the overall gain falls to 50. CO5

(i) Calculate the fraction of the output voltage feedback.

(ii) If this fraction is maintained, calculate the value of the amplifier gain required if the overall stage gain to be 75.

5\. a) What is Precision Rectifier? Explain Precision Rectifier with proper circuit diagram. CO5

OR

Show that the output is the integral of the input with an inversion and scale multiplier of 1/RC.

5\. b) Explain the diagram of Peak Detector circuit CO5

5\. c) Fig.5 (c) shows the OP-amp integrator and the square wave input. Find the output voltage and output wave shape. CO5

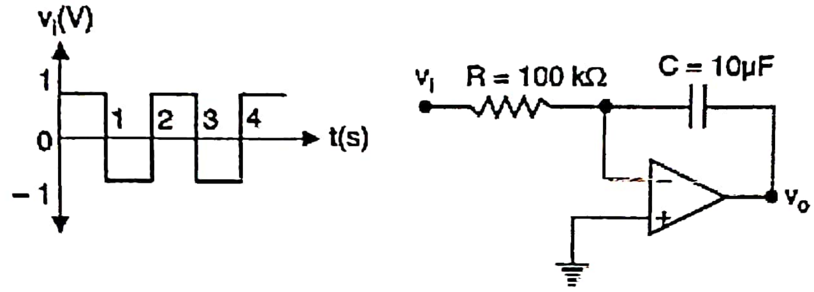

Fig. 5 (c)

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International Islamic University Chittagong

Department of Computer Science & Engineering

B. Sc. in CSE Semester Final Examination, Autumn-2023

Course Code: EEE-1221 **Course Title: Electronics**

Total marks: *50* Time: *2* hours *30* minutes

[Answer ***all*** the questions. Figures in the right hand margin indicate full marks. Separate answer script must be used for Group A and Group B]

Group-A

a) What is threshold voltage in MOSFETs? Describe the basic working principle of an enhancement-mode N-channel MOSFET. CLO4 U 5

OR

What is pinch off voltage? Discuss the operation of an n-channel JFET that has Pinch off voltage of 4V with neat diagrams.

b) Identify the differences between JFET and BJT? A JFET has a drain current of 5mA. If I_{DSS}= 10mA and V_{GS}(off) = -6V, find the value of (i) V_{GS} and (ii) V_P CLO4 An 5

a) What is a multivibrator in electronics? With neat diagram explain the operational principles of an multivibrator which produces square wave as output. CLO4 Ap 5

OR

Illustrate the concept of "saturation" and "cutoff" when referring to a transistor's switching behavior by interpreting the transistor Characteristics curve?

b) Fig-2(b) shows the transistor switching circuit. Given that R_B = 2.7 kΩ, V_{BB} = 2V, V_{BE} = 0.7V and V_{Knee} = 0.7V. CLO4 An 5
(i) Calculate the minimum value β for saturation.
(ii) If V_{BB} is changed to 1V and the transistor has minimum β =50, will the transistor be saturated.

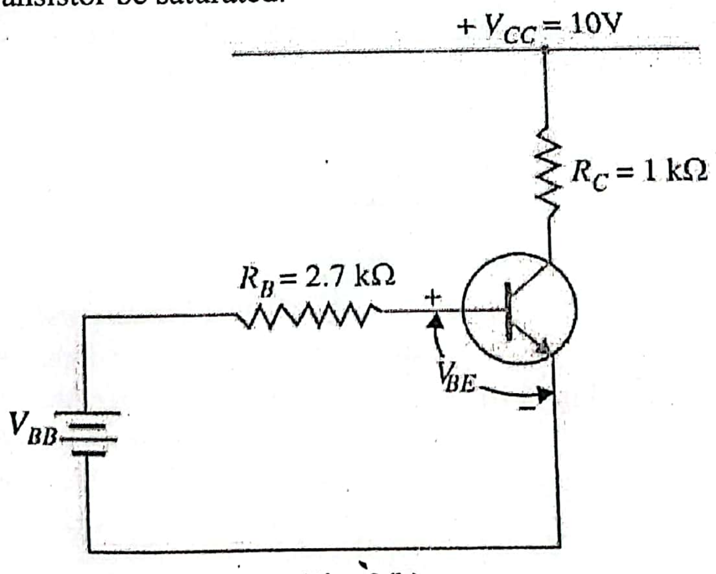

Fig. 2(b)

Group-B

3. a) Can you explain what an operational amplifier (op-amp) is, and could you derive the expression for voltage gain in an inverting amplifier?" CLO5

3. b) Illustrate the output voltage waveform with proper mathematical expression for the circuit given below CLO5

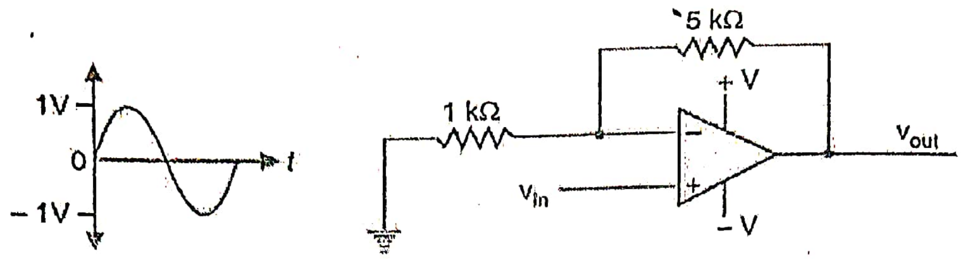

Fig. 3(b)

4. a) Can we use positive feedback to cancel noise? If not, which feedback is feasible for that purpose? Derive the negative voltage feedback gain (A_{vf}) of an operational amplifier. CLO5

OR,

How does a Differentiator amplifier configuration works using an op-amp? Derive the formula for calculating the gain in this configuration.

4. b) When negative voltage feedback is applied to an amplifier of gain 100 the overall gain falls to 50. CLO5
i) Calculate the fraction of the output voltage feedback.
ii) If this fraction is maintained, calculate the value of the amplifier gain required if the overall stage gain is to be 75.

5. a) Write a short note on comparator circuit and characteristics of it with proper diagram. CLO5

OR,

How is the peak detector used to find peaks in a signal? Elaborate the methodology.

5. b) Analyze the output voltage in the context of Fig. 5(b), where an op-amp integrator is connected to a square wave input. CLO5

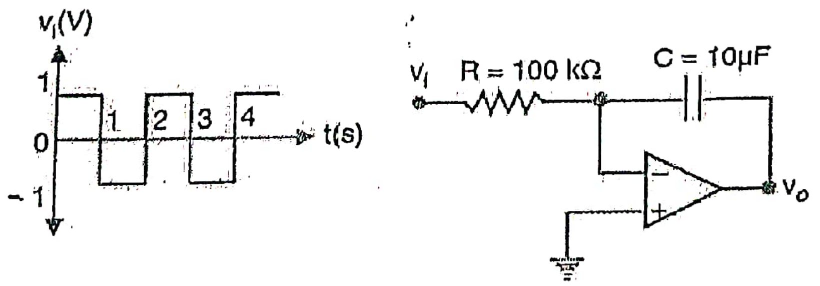

Fig: 5(b)

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International Islamic University Chittagong

Department of Computer Science & Engineering

B. Sc. in CSE Semester Final Examination, Spring-2024

Course Code:EEE-1221Course Title: Electronics

Total marks: *50* Time: *2* hours*30* minutes

[Answer ***all*** the questions. Figures in the right hand margin indicate full marks.
Separate answer script must be used for Group A and Group B]

Group-A

1. a) What is pinch-off voltage in MOSFETs? Explain the operation of n-channel enhancement mode MOSFET. CLO2 U 5

OR,

What is JFET? Draw the symbol of N-Channel JFET and P-Channel JFET. Describe the working principle of N-Channel JFET.

b) A JFET has a drain current of 5mA. If I_{DSS}=10 mA and $V_{GS(off)}$ = -6V, Find the value of i) V_{GS} and ii) V_P CLO3 An 5

2. a) How many types of switching circuits are there? Describe the switching action of a transistor, illustrating the 'OFF' region, 'ON' region, and 'Active' region on its output characteristics? CLO2 Ap 5

OR,

(a Design a Multivibrator circuit that can generate square wave output with no stable state. Explain its operation.

b) *Fig.* 2(b) shows the transistor switching circuit. Given that R_B = 2.7 kΩ, CLO3 An 5
V_{BB} = 2V, V_{BE} = 0.7V and V_{knee} = 0.7V.
  i) Calculate the minimum value of β for saturation.
  ii) If V_{BB} is changed to 1V and transistor has minimum β = 50, will the transistor be saturated.

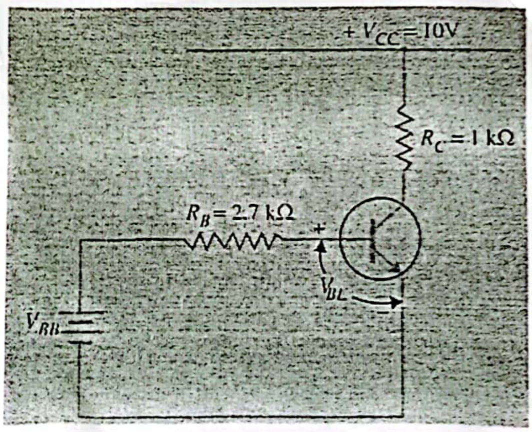

Fig. 2(b)

Group-B

3\. a) Sketch a neat diagram and derive an expression for the voltage gain of a non-inverting amplifier. CLO2 U 5

b) Find the output voltage with proper mathematical expression for the circuit given below CLO3 An 5

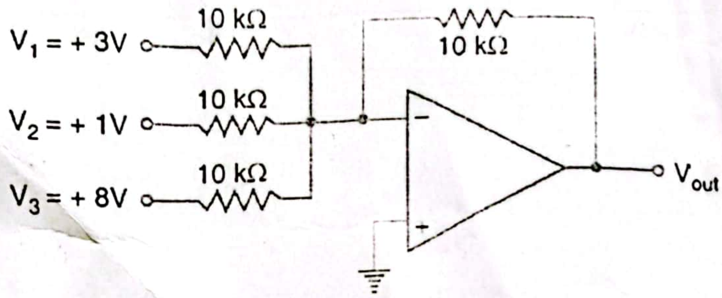

Fig. 3(b)

4\. a) What is negative feedback? Derive the gain of negative feedback in amplifier. CLO2 Ap 5

OR,

Show that the input impedance of amplifier increases due to negative voltage feedback.

b) When negative voltage feedback is applied to an amplifier of gain 100, the overall gain falls to 50. CLO3 An 5

(i) Calculate the fraction of the output voltage feedback

(ii) If this fraction is maintained, calculate the value of the amplifier gain required if the overall stage gain to be 75.

5\. a) What is Oscillator? Briefly explain damped and undamped oscillations with proper illustrations. CLO2 U

OR,

Briefly explain the working principle of Heartly oscillator with proper circuit diagram.

b) Explain the diagram of Peak Detector circuit in details. CLO3 U